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The Selection of Scientific Talent in the Allocation of Research Grants[1]

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Career grants are an important instrument for selecting and stimulating the next generation of leading researchers. Earlier research has mainly focused on the relation between past performance and success. In this study we investigate how the selection process takes place. More specifically, we investigate which quality dimensions (of the proposal, of the researcher and societal relevance) dominate. We also study which phases in the process (peer review, committee review, interview) are dominant in the evaluation process. Finally, we investigate whether differences between disciplines are visible. The analysis of our data set, consisting of the reviews of 898 grant applications, shows that talent has different dimensions and therefore is not obvious. The evaluation of talent was found to be contextual, although there were only small differences between disciplines. Unlike the interviews with the applicants, the external peer reviews hardly influence the decision-making on grant allocation. The notion of talent was found to be the least evident in the social sciences and humanities.
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Introduction

One of the key elements in the Europe 2020 strategy is a strong increase of Europe's research capacity (European Commission, 2010). Large investments in higher education over the last decades have resulted in a massive increase in the number of higher education graduates, and between 1998 and 2006, OECD countries showed a 40% increase in doctoral graduates (Auriol, 2010). At the same time, temporary positions are increasingly common in academia, reflecting the increase of project funding (Lepori *et al.*, 2007; van Steen, 2011). Consequently, academic career opportunities are by far outnumbered by young researchers who hope to establish an academic career (Huisman *et al.*, 2002; van Balen, 2010), leading to strong competition among researchers (de Grande *et al.*, 2010). Securing a personal career grant is increasingly a necessity for a successful academic career, as it shows that the applicant is perceived as a strong talent.

Universities and research councils aim at selecting the most talented young researchers, using explicit and also often implicit criteria (van den Besselaar and Leydesdorff, 2009). As the quality of the research system and the careers of researchers depend on these selection processes, it is important to understand how they function. In this paper we study the selection of scientific talent through career grants. We will show how the selection proceeds through the various phases, how consistent these phases are with each other, and which phases and criteria are decisive for the final selection. We will also study the differences between disciplinary domains and between the three grant schemes under study.

Theoretical Background

Although 'scientific excellence' and 'talent' are commonly used (Addis and Brouns, 2004), the meaning of these concepts is contested (Hemlin, 1993). Much debated is, for example, whether talent is innate or acquired. Talent has been explained by innate factors (e.g. Gross, 1993; Baron-Cohen, 1998), but this research is often criticized as mainly anecdotal and retrospective (Ericsson *et al.*, 2007). Talent is also conceived in terms of personality (and its genetic components), effecting scientific performance (e.g. Busse and Mansfield, 1984; Feist, 1998; Feist and Barron, 2003). However, others claim that people are not born to be a genius (Howe *et al.*, 1998), as excellence is mainly determined by environmental factors, including early experiences, training, preferences and opportunities. If that is the case, talent should not be considered as a quality in itself, but more as innate *potential*. It is a process that enhances training and performance and involves domain-specific expertise (Simonton, 2008). Consequently, it is difficult to decide who is a talented researcher and who is not.

Selection committee members review and discuss grant proposals or job applications and jointly identify the most excellent ones — often using peer review reports. This decision-making process entails among other things reference to one's expertise, explanation of preferences, discussion between proponents and opponents, obedience (or not) to procedures and rules, and finally reaching agreement. Lamont (2009) describes this type of evaluation as a social, emotional and interaction process. In an observation study of grant review panels, she shows that scientific excellence does not mean the same to everyone. Committee members from different fields, with a variety of motivations, use different criteria. And even within fields, people define excellence in various ways. As excellence is not the same for everyone, but subject to discussion and (dis)agreement, one may consider talent to be 'socially constructed' (Smith, 2001). More generally, social constructivism poses that scientific knowledge and the judgement thereof is constructed through interpretations, negotiations, and accidental events (Knorr-Cetina, 1981). On the other hand there is the Mertonian sociology of science, which prescribes how scientists should behave according to the norms and values of science, the so-called 'ethos of science' (Bornmann, 2008). One of these norms is *universalism*, which means that the judgement of knowledge claims should be based on scientific criteria only, without interference by personal or social backgrounds of the reviewed and reviewers (Merton, 1973 [1942]). Applied to talent selection, access to scientific careers should be based on scholarly competence alone.

Even within the Mertonian norms, grant applications (and job applicants) are not evaluated and selected separately, but in comparison to competing applications (Smith, 2001). Quality is socially and contextually defined from a specific point of reference that evolves during the evaluation process (Lamont, 2009). As a result of this *contextual ranking*, one may expect that the same grant application can be valued differently across committees, process phases, and time. This is exactly what Cole *et al.* (1981) found in their study on the reviewing of applications for research grants from the National Science Foundation (NSF). After reviewing all and selecting half of the applications, a second group of peers reviewed and ranked the same set again. The two rankings differed substantially. Several proposals that were rejected by the NSF would have been granted if the selection had been based on the second ranking. What then determines whether a proposal is evaluated to be more excellent than the other? How is talent selected within review committees?

Engaging peers is essential, as they are best suited to review the work of 'colleagues' within their specialty (Eisenhart, 2002). However, peers are often close to the applicants, and this creates tension between peer expertise and impartiality (Eisenhart, 2002; Langfeldt and Kyvik, 2011). This relates to another tension: peer reviews ought to be neutral, but not scholarly neutral. Personal interests should be eliminated and the evaluation should be based on scholarly discretion. But where are the boundaries? A third tension exists between unanimity and divergence. Grant review committees are expected to reach a unanimous decision, but at the same time divergence is considered of great value. Divergent assessments lead to discussion and contribute to the dynamics of science (Langfeldt and Kyvik, 2011). As scientific excellence is not unambiguous, but defined by reviewers and committee members in their own way, grant allocation clearly is a dynamic process.

Earlier studies on selection of applications focused mainly on *past performance* of the applicant.[2] Melin and Danell (2006) compared the past performance of successful and just unsuccessful applicants to the Swedish Foundation for Strategic Research. As the mean number of publications only slightly differed between the two groups, the applicants awarded grants can hardly be considered to be more productive than the rejected applicants. A study of the past performance of grant applicants in the Netherlands did find the expected difference in track record between awarded and all rejected applicants (van den Besselaar and Leydesdorff, 2007; van den Besselaar and Leydesdorff, 2009). However, comparing the past performance in terms of publications and citations of the awardees with the most successful rejected applicants, the latter have a slightly better average past performance than the awarded applicants. A later study found the same for German career grants (Hornbostel *et al.*, 2009) and for international career grants in molecular biology (Bornmann *et al.*, 2010). In their classical study on reviews of grant applications at the NSF, Cole *et al.* (1981) found a weak correlation between past performance and funding granted, concluding that the allocation of grants seems to be determined about half by characteristics of the applicant and the proposal, and about half by chance. Other research showed academic rank (Cole *et al.*, 1981), research field (Laudel, 2006), type of research (Porter and Rossini, 1985), and academic and departmental status (Cole *et al.*, 1981; Bazeley, 1998; Jayasinghe *et al.*, 2003; Viner *et al.*, 2004) (weakly) correlate with quality assessment of the application or applicant.

The chance element reported by Cole *et al.* (1981) can partly be ascribed to the subjective character of the reviewing process and the social construction of scientific quality. According to Lamont (2009), it is impossible to completely eliminate this subjectivity, as it is simply the nature of the processes that is intersubjective. The outcomes of the review process therefore are affected by who is conducting the review and how the committee is composed (Langfeldt, 2001; Eisenhart, 2002; Langfeldt and Kyvik, 2011). Different mechanisms can be discerned. Firstly, committee members who are nominated by the applicants, give higher ratings (Marsh *et al.*, 2008). Secondly, relations between reviewers and applicants influence the ratings. Researchers affiliated with reviewers received better reviews than those without this type of affiliation (Sandstrom and Hallsten, 2008). Finally, the way the review process is organized, influences the outcomes (Langfeldt, 2001).

Most research on grant allocation focuses on the outcomes, searching for predictors for success. The internal selection mechanism hardly has been studied, and we therefore do not know what happens during the selection process (Bornmann *et al.*, 2010). Only few studies have been conducted on the individual steps of the selection process (e.g. Hodgson, 1995; Bornmann *et al.*, 2008). Bornmann *et al.* (2008) applied a latent Markov model to grant peer review of doctoral and post-doctoral fellowships. Their model shows that the first stage of the selection procedure, the external reviewing, is of great importance for the final selection decisions. External reviews had to be positive for fellowship applicants to have a chance of being approved. However, van den Besselaar and Leydesdorff (2009), using a different method, could not confirm this. And no correlation was found between the decision and the external review score within the top 50% of the applicants.

In this paper we will analyse quality evaluation by looking into the dynamics of the selection process itself. We will study the role of the different talent dimensions and of the different phases of the selection process, and how this influences the final outcome.

The Case

Our data set consists of 1,539 career grant applications. These involve personal grants for researchers in three different phases of their careers:

- The early career grant scheme (ECG) for researchers who received a Ph.D. within the previous three years. The grant offers them the opportunity to develop their ideas further.
- The intermediate career grant scheme (ICG) for researchers who completed their doctorates within a maximum of 8 years and who have already spent some years conducting post-doctoral research. The grant allows them to develop their own innovative research line and to appoint one or more researchers to assist them.
- The advanced career grants scheme (ACG) for senior researchers with up to 15 years post-doctoral experience, and who have shown the ability to successfully develop their own innovative lines of research and to act as coaches for young researchers. The grant allows them to build their own research group.

Figure 1 briefly describes the selection procedure. If the number of applications in the ECG and ICG programme is over four times higher than the number of applications that can be awarded (as generally is the case), a pre-selection will take place — which resulted in our case in a rejection of about 40% of the applications. Because our data set contains no further information on the criteria and assessments in the pre-selection, we do not include this phase in our study. In the ACG programme, researchers first submit a pre-proposal. The selected applicants are invited to submit a fuller, more detailed proposal. The selection of pre-proposals is also left out from our study, for the same reasons. This reduces the data set to 897 applications.

Next the applications are sent to external referees, considered to be experts in the field of the applicant. The number of referees varies between two and six per proposal. The reviews and the applicants' rebuttal are sent to the review

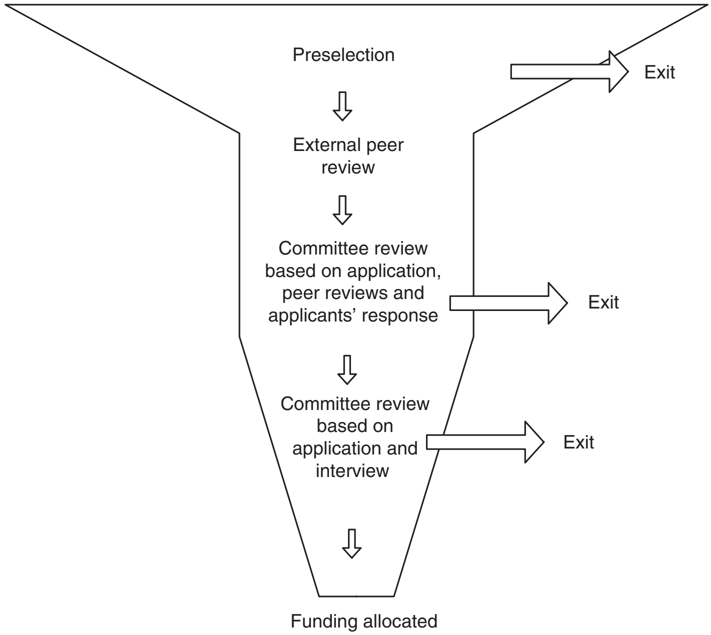

Figure 1. The general grant allocation procedure.

committee. Partly based on this input, the committee evaluates every proposal on three criteria: quality of the researcher (QR), quality of the proposal[3] (QP), and research impact (RI).[4] The score on research impact is only taken into account if it is better than the proposal score.[5] When this is the case (QP < RI), the final committee score is calculated as follows:

$$\text{Total committee score} = \frac{1}{2}QR + \frac{1}{4}QP + \frac{1}{4}RI$$

If the research impact is scored lower than the quality of the proposal (If QP > RI), the committee score is calculated as:

$$\text{Total committee score} = \frac{1}{2}QR + \frac{1}{2}QP$$

The total committee score leads to a ranking of the applications, which determines who proceeds to the next round: the interview, where the applicants present their proposal for the committee.

Hereafter the committee again evaluates every interviewed applicant (N=552) on the same three criteria, taking into account the information from

Table 1 Number of applications per scientific domain and funding programme across the selection procedure

	ECG			*ICG*			*ACG*		
	1st review[a]	*2nd review*[b]	*Granted*	*1st review*	*2nd review*	*Granted*	*1st review*	*2nd review*	*Granted*
SSH									
N	141	129	54	111	70	28	22	22	9
%		91.5	38.3[c]		63.1	25.2		100.0	40.9
STE									
N	151	70	40	124	65	33	34	34	12
%		46.4	26.5		52.4	26.6		100.0	35.3
LMS									
N	161	76	49	118	56	28	35	30	10
%		47.2	30.4		47.5	23.7		85.7	28.6
Total									
N	453	275	143	353	191	89	91	86	31
%		60.7	31.6		54.1	25.2		94.5	34.1

[a]External reviewers & first committee review.
[b]Second committee review.
[c]If we include all applications, also those rejected in the pre-selection phase, the SSH success rate is lower than the two others. This is due to the very high rejection rate in the pre-selection for social science applications.

the previous phases. To arrive at the final committee score, the same calculation rule is used as prior to the interview. The ranking of the final committee scores determines which applications will receive funding and which are rejected.

The research council consists of eight scientific divisions,[6] which are aggregated into three domains[7]: (1) Social Sciences and Humanities (SSH), (2) Science, Technology and Engineering (STE), and (3) Life and Medical Sciences (LMS). In our analyses we will distinguish between these domains when relevant. Table 1 gives an overview of the number of applications per programme and domain. As mentioned earlier we do not include the applications rejected in the pre-selection phase.

Our data include several attributes of the applications and applicants: gender, the grant scheme, the scientific division and the domain of the application, the referee scores, the committee scores on the three criteria, and the decisions. Between a third (ACG) to a quarter (ICG) of the applications that made it through the pre-selection, received funding (Table 1).

Research Questions

The grant allocation procedure (Figure 1) resembles a pipeline model. At the start there is a big pool of applicants, but as the procedure progresses the number of applicants decreases, with only a minority successfully reaching the end: obtaining of funding. In this study we aim to understand how applications pass the selection procedure and what determines which applications are eventually successful and which are expelled along the way. This should show how excellent talents are identified or created by the selection process. We answer the following research questions:

(1) *How evident is talent?*
How strong is the correlation between reviewers' scores, and between the committee scores and reviewers' scores? The stronger they correlate, the more 'evident' talent is. Secondly, do scores differentiate? Do the selected applicants have significantly higher scores than the non-selected?

(2) *Is talent selection dependent on the procedure?*
Do the rankings of applications in the different phases of the procedure correlate? Is the result stable, or does additional information in later phases result in strong fluctuations? And, are reviewers using the evaluation scales consistently through the procedure — do scores have a stable meaning?

(3) *Does talent have various dimensions?*
Do the three main criteria used by the committees represent different dimensions — or do they in fact measure the same? If they are different, are the rankings dependent on weighting the dimensions?

(4) *Which phases of the process and which criteria eventually determine which applicants are considered to be talents*?
A logistic regression analysis is used to identify which criteria and phases of the selection procedure have most influence on the final grant allocation decision.
After answering these questions, we will discuss the implications of the findings for the system of selecting and granting research proposals.

Methods

Using ANOVA, we first compare the review scores per domain (SSH, STE and LMS) and per funding programme (ECG, ICG and ACG) in order to determine whether evaluation practices differ between scientific domains and career phases. Some of the following analyses are conducted on domain and programme level, others on the complete data set. In the latter case the data are standardized beforehand on the domain and programme variables.

Agreement between reviewers is analysed by calculating the standard deviation and the maximum difference between review scores per application and by rank order correlation. We will rank the review scores per step of the selection process and compare these rankings to see if applicants were evaluated differently at various moments of the procedure. The use of the evaluation scale is analysed with Chi-square tests. Rank order correlations are calculated between the three evaluation criteria used by the committees. This will show whether talent has one or various dimensions. Finally, to identify the predictors for talent selection we conducted multiple logistic regression analysis.

Results

Evaluation practices differ between the scientific domains, as Table 2 indicates. Please note that *lower numbers* represent *better scores*. The SSH applications have the best committee scores on almost all criteria, but low referee scores. The LMS applications have the lowest referee scores and the lowest average committee scores before the interview. Finally, STE applications have the lowest average committee scores after the interview, but the best referee scores. Because of these (although not always significant) domain-related differences, we will distinguish between the three domains in our analysis.

We also check for differences between the funding programmes, as evaluation practices may differ here too. Table 3 shows that in the ECG scheme the lowest scores are assigned on almost all criteria and that the highest scores are given in the ACG scheme. Therefore we will also distinguish between the three funding programmes in our analyses.

Table 2 Average evaluation scores per domain and ANOVA results

	External referees	*Committee review before interview*				*Committee review after interview*			
		Quality researcher	*Quality proposal*	*Research impact*	*Committee average*	*Quality researcher*	*Quality proposal*	*Research impact*	*Committee average*
SSH	2.60	*2.50*	*3.03*	*3.19*	*2.75*	*2.55*	3.14	*3.09*	2.84
STE	*2.38*	2.84	3.30	3.48	3.02	**2.64**	**3.15**	**3.44**	**2.84**
LMS	**2.64**	**3.02**	**3.54**	**3.76**	**3.22**	2.62	*3.00*	3.15	*2.76*
F	6.31**	21.02*	13.87*	10.61*	16.49*	0.38	0.88	3.84**	0.31

*$p<0.001$, **$p<0.05$.
Italic = best average score; **bold** = lowest average score.

Table 3 Average evaluation scores per funding programme and ANOVA results

	External referees	*Committee review before interview*				*Committee review after interview*			
		Quality researcher	*Quality proposal*	*Research impact*	*Committee average*	*Quality researcher*	*Quality proposal*	*Research impact*	*Committee average*
ECG	**2.71**	**2.91**	**3.45**	3.53	**3.13**	**2.76**	**3.28**	*3.14*	**2.98**
ICG	2.44	2.78	3.23	**3.63**	2.97	2.55	3.00	**3.39**	2.74
ACG	*2.12*	*2.28*	*2.80*	*3.31*	*2.51*	*2.20*	*2.75*	3.17	*2.44*
F	18.28*	15.42*	12.41*	2.27	14.82*	9.72*	7.71*	2.24	9.17*

*$p<0.001$.
Italic = best average score; **bold** = lowest average score.

Table 4 Disagreement in evaluations by external referees per domain and funding programme

	Early career grant		*Intermediate career grant*		*Advanced career grant*	
	Maximum disagreement[a]	*Average disagreement*[b]	*Maximum disagreement*[a]	*Average disagreement*[b]	*Maximum disagreement*[a]	*Average disagreement*[b]
All	1.57	1.05	2.06	1.10	2.22	1.06
SSH	1.60	1.13	2.25	1.21	2.75	1.28
STE	1.68	1.09	1.76	0.95	2.03	0.95
LMS	1.45	0.94	2.18	1.16	2.08	1.02

[a]Mean of maximum difference between review scores per application.
[b]Mean of standard deviation review scores per application.

The evidence of talent

The applications are reviewed by external reviewers and (twice) by a council committee. The number of external reviewers per proposal varies between two and six.[8] In general there are two reviewers for the ECG, three for the ICG and four for the ACG. The external reviewers assign scores from 1 (highest) to 6 (lowest). We calculated the difference between the maximum and minimum review score per proposal. As Table 4 shows, the reviewers disagree least in the ECG scheme (M=1,59; SD=1,27) and most in the ACG scheme (M=2,22; SD=1,33). The level of disagreement differs significantly between the schemes, $F(2, 895)$=18,72, $p<0.001$, indicating that the further an applicant is in his career, the stronger the average disagreement about his quality.

Taking into account that the number of reviewers varies per grant scheme, we compare the average distribution of review scores per proposal (mean standard deviation, Table 4). The standard deviation can range from 0 (if all reviewers totally agree) to 3.54 (when reviewers totally disagree). However, no significant difference was found between the programs. Although the maximum disagreement between reviewers increases with the career phases, the mean disagreement remains the same. The higher number of reviewers in the IGC and ACG scheme explains this: The more reviewers per proposal, the smaller the weight of reviews with extreme scores.

We repeated the analysis for each of the domains, to find out whether agreement on talent differs between the domains. Only in the ICG scheme the average disagreement (standard deviation) between reviewers significantly differs between the domains ($F(2,351)=5.25$, $p<0.01$). In the ECG and ACG schemes no significant differences were found. Finally, in all career phases the reviewers in the Social Sciences and Humanities seem to disagree stronger than in the other domains.

The selection of interview candidates is carried out by a committee, taking into account the external reviews and the applicants' rebuttal. The correlation between the standardized external review scores and the committee reviews is used to determine the extent to which evaluators in different phases of the procedure agree on the quality of applicants. In all domains the external reviews correlate moderately strong (ECG and ACG: $\tau=0.53$, $p<0.001$; ICG: $\tau=0.52$, $p<0.001$) with the first committee scores.[9] After the interview, the same committee evaluates the applicants again. The correlation between the committee scores before and after the interview is also moderately strong in the domains of STE and LMS ($\tau=0.42$, $p<0.001$) and strong in SSH ($\tau=0.62$, $p<0.001$).

The average scores are used to distinguish between the talented and the less talented applicants, but to what extent do these scores discriminate? We ranked (for the complete set and per domain) all applications using the standardized average review score. As Figure 2 shows for the complete set, the distribution has no clear-cut off point, and a similar pattern exists at domain and programme level. The dotted line indicates the *de facto* cut-off point for applications selected for the next (interview) phase. However, this selection

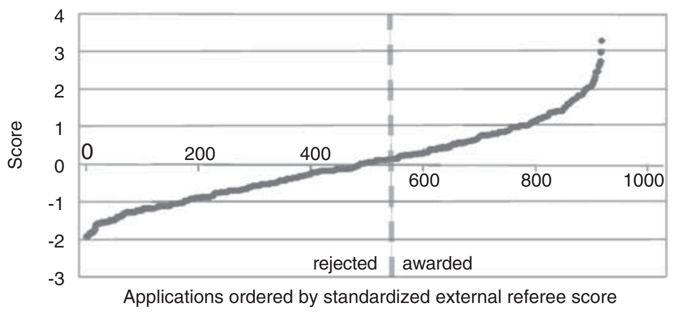

Figure 2. Standardized external referee scores for the complete set of applications.

boundary does not follow the scores, as the difference between success and no success is very small. Similar patterns were found for the committee scores, where the difference between success and failure is very small too.

To conclude, no clear 'boundaries of excellence' could be identified. Moreover, the average scores in the three phases of the procedure only correlate moderately strongly, and that may result in considerable changes in the rankings. This issue will be addressed in the next section.

Effects of the procedure

The selection procedure includes three evaluation phases in which new information is added which may influence the resulting assessment. Figures 3 and 4

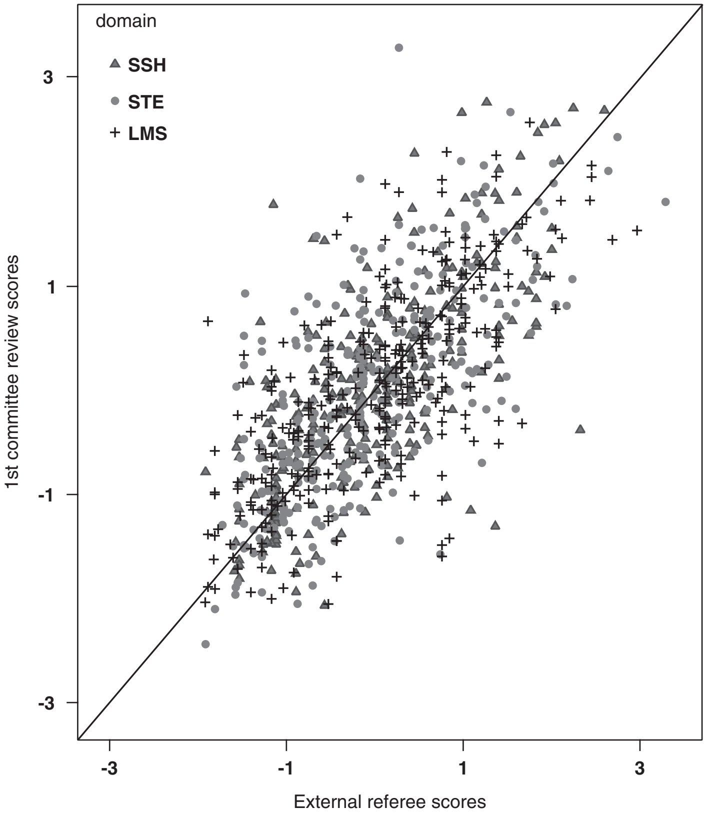

Figure 3. First committee review by external referee score.

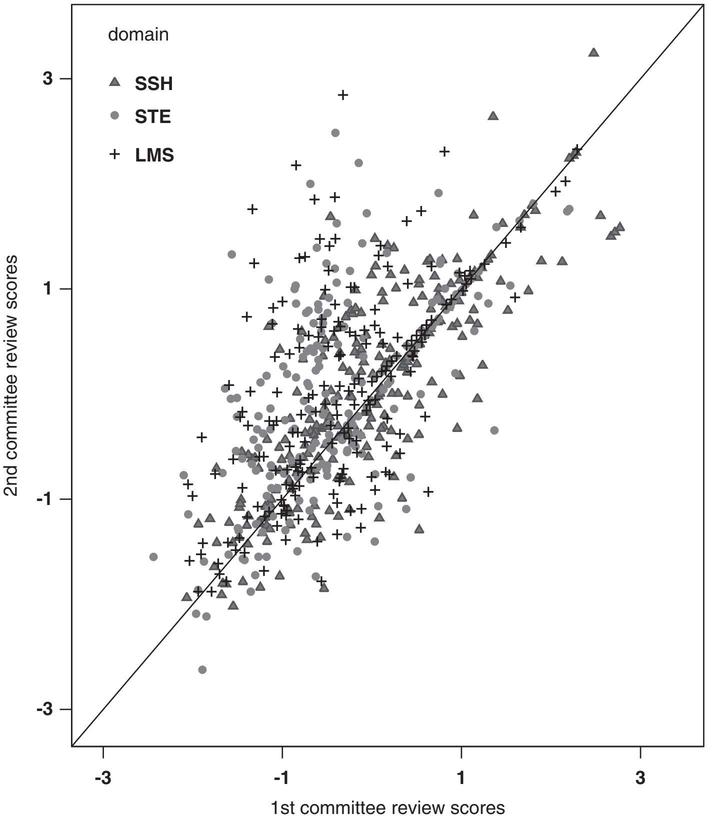

Figure 4. Second committee review by first committee review.

show how applications are evaluated differently at different moments of the procedure, based on the standardized review scores. Left of the diagonal in Figure 3 are the applications that had a better (=lower) external review score than a first committee score. On the right side are the applications that had a better (=lower) committee score than review score. Clearly, the scores and the relative position of applications changes during the procedure. If external (peer) review scores would have been leading, the set of applicants invited to the interview would have been rather different. Since both evaluations are based on about the same information, this implies that talent evaluation depends on the way it is organized — it is 'contextual'.

In Figure 4, the committee reviews before and after the interview are compared, with right from the diagonal those applications that score higher after the interview than before, whereas left of the diagonal the opposite is the case. Committees adjust their assessments after the interview, and quite some applicants score much higher after than before the interview and vice versa.

This implies that if grant allocation had been based on the evaluation scores before the interview, the outcome would have been different. How strong is this effect? To answer that question, we compare the rankings of applications between the three evaluation moments, showing the importance of the various phases of the selection process. Differentiating between the funding programmes and scientific domains, we firstly determine how many applications would not have been selected for the interview when the selection would have been based on the external review scores (Table 5).

In the ECG and ICG programmes, 27 and 21 applicants respectively were not selected for the interview phase, although they were among the highest external referee scores. Consequently, in both programmes about 17% of the selected applicants would not have been selected if the external referee scores

Table 5 Applications that would not have been selected for the interview — if specific scores would have been used (per domain and funding programme)

Funding programme	*Domain*	*N selected for interview*	*External referee score*	*1st committee review score*[a]
ECG	SSH[b]			
	N	8	1	0
	%		12.5	0
	STE			
	N	70	14	3
	%		20.0	4.3
	LMS			
	N	76	12	10
	%		15.8	13.2
ICG	SSH[b]			
	N	27	8	5
	%		29.6	18.5
	STE[b]			
	N	36	4	1
	%		11.1	2.8
	LMS			
	N	60	9	5
	%		15.0	8.3

[a]Actually this means that the committees do not follow the decision-making rules and have an in fact additional autonomy in decision-making. The arguments used were not in the data set. This we hope to study in a follow-up project.
[b]In some divisions within this domain all applicants were invited for the interview; these are excluded from this table.

had been paramount. In the ECG programme, the external referee scores contribute least to the interview selection in the STE domain, 20% of the decisions would have been different if these scores would have been decisive. In the ICG scheme this holds for the SSH domain, almost 30% of the decisions would have been different if these scores would have been decisive. In the STE domain, the committee strongly follows the external reviews, as almost 90% of the decisions are in line with the external referee scores.

According to the procedure, the committee score is decisive. However, about 8% of the successful early career applicants and 9% of the successful intermediate career applicants were not selected despite they were among the highest total committee scores. There were respectively 13 and 11 rejected applicants with a higher total committee score than the selected applicants. Distinguishing between the domains shows that in the ECG the deviation from the prescribed procedure is largest in LMS (13%). In ICG almost one out of five selection decisions in the SSH domain is not based on total committee scores. In STE all decisions but one followed the decision-making rules.

Grant allocation is the final step in the selection procedure. Table 6 displays the number of applications that would not have been granted if the allocation had been based on respectively (1) the external referee scores, (2) the first committee scores (before the interview) and (3) the second committee scores (after the interview). If the grant allocation had been based entirely on the evaluation by the external referees, 25% of the ECG applicants to 29% of the ACG applicants, with the ICG applicants in between (27%), would not have been allocated a grant. In the ACG we see that this percentage is even much higher for the domains of SSH and STE, 44% and 33% respectively. In

Table 6 Applications (*N*, %) that would not have been granted based on specific scores (per domain and funding programme)

Programme	*Domain*	*Granted*	*External review score*	*1st committee score*	*2nd committee score*[a]
ECG	SSH	54	10 (18.5%)	10 (18.5%0	1 (1.9%)
	STE	40	13 (32.5%)	13 (32.5%)	0 (0%)
	LMS	49	13 (26.5%)	12 (24.5%)	2 (4.1%)
ICG	SSH	28	8 (28.6%)	5 (17.9%)	0 (0%)
	STE	33	9 (27.3%)	7 (21.2%)	0 (0%)
	LMS	28	7 (25.0%)	4 (14.3%)	1 (3.6%)
ACG	SSH	9	4 (44.4%0	3 (33.3%)	0 (0%)
	STE	12	4 (33.3%)	3 (25.0%)	0 (0%)
	LMS	10	1 (10.0%)	2 (20.0%)	0 (0%)

[a]Actually this means that the committees do not follow the decision-making rules and have an in fact additional autonomy in decision-making. The arguments used were not in the data set. This we hope to study in a follow-up project.

the LMS domain it is much lower (10%), as only one rejected application would have been granted based on the external referee scores.

If interviews would not have been part of the procedure — and this is the case in many funding schemes — and the first committee reviews would have determined the grant allocation, about a quarter of the grants in the ECG and ACG schemes and a fifth in the ICG scheme would have been allocated to currently unsuccessful applicants. These deviations differ between domain-programme combinations, ranging from 14% in LMS (ICG) to 33% in STE (ECG) and in SSH (ACG), but no pattern could be identified. These findings imply that the interview considerably changes the assessment of talent. We hope to investigate the dynamics, the criteria (implicitly) applied, and the effects of the interview in a follow-up study. As the last column of Table 6 shows, the final decision largely corresponds to the second committee score, as the rules prescribe.

What do the scores represent?

After showing how the *perception of talent* did change, we will now study changes in the *use of the evaluation scale* (as distinct from the evaluation of the applications). The six-point scale ranges from excellent (1), very good (2), and very good/good (3), to good (4), fair (5) and poor (6), clearly an 'absolute scale'. The committee members assign a score between 1 and 6 to each application, based on three criteria (quality researcher, quality proposal and research impact). Table 7 shows the mean scores and standard deviations for two typical evaluation committees, before and after the decision about which applicants are invited for an interview.

In case 1, about 50% of the highest scoring applications were selected. As expected, the means for all applicants (first review, all applications) are

Table 7 Use of evaluation scale

Case	*Programme & domain*	*1st committee review all applications*			*1st committee review selected applications*			*2nd committee review selected applications*		
		Researcher	*Proposal*	*Total*	*Researcher*	*Proposal*	*Total*	*Researcher*	*Proposal*	*Total*
1	ECG-STE									
	N	34	34	34	18	18	18	18	18	18
	Mean	2.89	3.43	3.10	2.28	2.87	2.55	2.56	3.23	2.84
	SD	0.84	0.78	0.73	0.54	0.45	0.47	0.83	0.95	0.84
2	ICG-SSH									
	N	34	34	34	34	34	34	34	34	34
	Mean	1.69	2.18	1.91	1.69	2.18	1.91	1.79	2.20	1.98
	SD	0.38	0.43	0.35	0.38	0.43	0.35	0.38	0.49	0.39

Table 8 Changing use of the scores by changing context

	Reduction of nr applicants after 1st committee evaluation	
	Yes[a]	*No*
*Decrease average score**		
No	4 (28.6%)	10 (83.3%)
Yes[b]	10 (71.4%)	2 (16.7%)
*Increase standard deviation***		
No	4 (28.6%)	8 (66.7%)
Yes[b]	10 (71.4%)	4 (33.3%)
Total	14 (100%)	12 (100%)

[a]Yes = changing context.
[b]Yes = using the score values in a relative way.
$N = 26$.
*$X^2 = 7.797$, $p = 0.005$; **$X^2 = 3.773$, $p = 0.05$.

lower than the means for selected applicants only (first review selected applicants).[10] The standard deviation of the whole set of applicants is larger than for the selected only — indicating an expected smaller variation among the selected applicants. However, average and standard deviation of the scores *after* the interview (second committee score) are equal to the values for all applicants in the first review, suggesting that the committee again has applied *the whole scale*: some of the applications scoring very good and excellent in the first round are now only fair or even poor. In this case, the scale is used in a *relative* way, and not as an *absolute* one. In case 2 no selection took place, as all applicants were interviewed. The interview did influence individual scores, but the average and the standard deviation before and after the interview remain about the same. No changes in the use of the scale seem to have occurred here.

Comparing the 14 ‘selective’ committees with the 12 ‘non-selective’ committees (in Table 8) shows a significant correlation between the change of context (selection between the phases or not) and the change of the use of the scale (relative or absolute scale). Consequently, the assessment of talent depends upon the context, on the procedure: for example, an interview, as showed in the previous section, and the number of competitors, as showed in this section.

The dimensions of talent

Earlier we showed that the external reviews correlated moderately strongly with the committee reviews. Distinguishing between the three criteria used by

Table 9 Correlations between the standardized committee review scores for the three criteria per domain

	SSH		*STE*		*LMS*	
	QR	*QP*	*QR*	*QP*	*QR*	*QP*
Before interview						
QP	0.50*		0.50*		0.44*	
RI	0.33*	0.41*	0.38*	0.49*	0.37*	0.47*
After interview						
QP	0.57*		0.56*		0.59*	
RI	0.37*	0.49*	0.31*	0.44*	0.41*	0.47*

*$p<0.001$.
QR = quality researcher; QP = quality proposal; RI = Research impact.

the committee shows that this moderate correlation is mainly due to a relatively weak correlation between external reviews and the committee scores for research impact, τ=0.22, $p<0.001$ (SSH); τ=0.29, $p<0.001$ (STE); τ=0.36, $p<0.001$ (LMS). In the LMS domain however, the external referee scores correlate even weaker with the committee scores for the researcher, τ=0.32, $p<0.001$. The external reviews are strongest related to the committee scores for the proposal, τ=0.55, $p<0.001$ (SSH); τ=0.55 $p<0.001$ (STE); τ=0.64, $p<0.001$ (LMS).

The three criteria are found to correlate moderately with each other (Table 9). Research impact correlates weakest to the quality of the researcher in all domains both before and after the interview, ranging from τ=0.31 to 0.41. The correlation between the quality of the proposal and the quality of the researcher increased after the interview in all domains, being strongest in LMS, from τ=0.44 before the interview to τ=0.59 after the interview.

This suggests that the three criteria represent different dimensions. The total score of the committee (as calculated following formulas from the method section) therefore depends on the weighting attributed to the different dimensions. This may change with the decision-making context. For example, the increasing emphasis on the applicability of research is reflected by first the introduction of the research impact score and by a gradual increase of its weight in the procedures over time.

Predictors for talent selection

The first decision is when committees select and reject applications for the interview round, based on the external reviews, the applicants' responses to these reviews, and the committees' own scoring on three criteria. In order to determine which of these variables best predict whether an application will be selected for the interview, we conducted a stepwise logistic regression analysis, including the average external referee score and the three committee scores.[11]

The model with only the external reviews predicts at 69.1% the cases correctly who go to the interview, slightly above the random correct prediction of 61.5%. In the full model, only the committee scores for the quality of the proposal and the researcher's quality are included, whereas the other variables are excluded (Table 10). These two variables predict in 77.3% of the cases correctly whether a researcher was invited for the interview or not.

After the interviews, the committee again scores the applications on the three criteria. A logistic regression analysis was done to predict the allocation decisions from the external referee scores and the three committee scores (Table 11). Again, external referee scores and the research impact scores do not contribute to the prediction. The committee scores for the proposal and for the researcher result in a correct classification of 83.1% of the cases. The model with only the external reviews predicts at 65.2% the cases who correctly receive funding, slightly above the random correct prediction of 52.3%.

Table 10 Logistic regression of the selection of interview candidates

	B (SE)	*X^2 (d.f.)*	*Nagelkerke R^2*	*Correct (%)*
Constant	−0.61* (0.10)			
Quality researcher	0.71* (0.13)			
Quality proposal	1.36* (0.15)			
Model		283.96* (2)	0.48	77.3
Not included				
External reviews	*0.23 (0.16)*			
Research impact	*0.15 (0.14)*			

*$p<0.001$.

Table 11 Logistic regression analysis to predict grant allocation decisions

	B (SE)	*X^2 (d.f.)*	*Nagelkerke R^2*	*Correct (%)*
Constant	0.46* (0.15)			
Quality researcher	1.40* (0.23)			
Quality proposal	1.80* (0.23)			
Model		294.97* (2)	0.65	83.1
Not included				
External reviews	*0.21 (0.18)*			
Research impact	*0.08 (0.19)*			

*$p<0.001$.

As we found rather modest differences in the evaluation practices between domains, we only differentiate between the three funding programmes when determining the importance of the evaluation criteria in different career phases. In all three programmes the external peer review score and the research impact score both do not predict the final funding decision and are removed from the model. The decision to grant an application is mainly based on the quality of the proposal and the quality of the researcher. Actually, the two criteria fairly well predict success (above 90%), although this is slightly less the case in the early career scheme (82.5%). This means that in decision making about early career grants, other factors play a bigger role.

However, the *de facto* weight of both criteria are found to differ between the funding programmes (Table 12). For the early career researchers the evaluation

Table 12 Logistic regression analysis to predict grant allocation decisions per funding programme

	B (SE)	*X^2 (d.f.)*	*Nagelkerke R^2*	*Correct (%)*
ECG				
Constant	−0.31* (0.20)			
Quality researcher	1.36* (0.32)			
Quality proposal	1.34* (0.32			
Model		120.26* (2)	0.59	82.5
Not included				
External reviews	*−0.30 (0.25)*			
Research impact	*0.02 (0.24)*			
ICG				
Constant	1.55* (0.39)			
Quality researcher	1.95* (0.51)			
Quality proposal	2.74* (0.55)			
Model		123.05* (2)	0.74	91.1
Not included				
External reviews	*0.41 (0.37)*			
Research impact	*−0.08 (0.38)*			
ACG				
Constant	3.96* (1.45)			
Quality researcher	4.86* (1.84)			
Quality proposal	5.63* (1.99)			
Model		81.85* (2)	0.89	93.7
Not included				
External reviews	*−0.19 (0.69)*			
Research impact	*−0.04 (1.14)*			

*$p<0.001$.

of the proposal and the researcher almost evenly determine the final selection decision, whereas for the intermediate and advanced career researchers the quality of the proposal is more important than the quality of the researcher.

Before formulating our conclusions, it is worthwhile mentioning some of the limitations of this study. Firstly, it is based on mostly but not all quantitative scores the applications received, as we did not have the individual committee member scores and the scoring in the pre-selection. Secondly, access to other data, such as reports of committee meetings, opinions of committee members and observations of committee meetings, would have helped to deeper grasp the dynamics of decision-making. Through that one may identify the (often implicit and unconscious) criteria used by committee members when reaching agreement on grant allocation. This we hope to do in a follow-up study.

Conclusions and Discussion

First of all, the moderate correlations between the criteria indicate that talent has different dimensions. This implies that the weight of the criteria strongly influences the selection process. For example, the weight of research impact is very low in the case we studied, but this is currently in a phase of change. Consequently, in the future other types of applicants may be selected as the most excellent talents.

Secondly, the scores change considerably between the phases. Some applicants, top ranked by the external referees, are not even invited for an interview by the committees. And these same committees regularly change their evaluation of applicants radically after the interview. The interview seems decisive, but how this works needs further investigation. Does the interview bring new information, leading to a different evaluation? In that case the procedure does influence the outcome considerably. Shouldn't the many existing procedures without interviews be abandoned? Or is it because other aspects of talent (such as communicative skills) and several cognitive, motivational and social processes (Lamont, 2009) play a role during the interview, as well as various psychological factors (Hemlin, 2009)?

Thirdly, the role of the external peer review in the quality assessment seems modest (Langfeldt *et al.*, 2010). Using only external review scores as predictor, the percentage of correctly predicted applications is only slightly higher than random (65.2% *vs* 52.3%), much lower than for the two other predictions (83.1%). Combined with the moderately (but not very) high correlation (τ=0.52) between reviewers score and committee score, this suggests that the committee takes the review scores into account, but not very strong.[12] Further study is needed, to reveal whether and how committee members value and use the peer review reports.

Fourthly, reviewers disagree, and the further a researcher is in his/her career, the more the reviewers disagree. In line with earlier studies, consensus about quality is lower in the social sciences and humanities than in sciences, technical sciences and life sciences (Cicchetti, 1991; Simonton, 2006). Committees and external reviewers also do not draw a clear line between talented and less-talented researchers, as very small differences in scores decide who receives a grant and who does not. As the funding decisions are of great importance for the careers of (especially) young researchers, career becomes partly a question of luck.

Fifth, we found only small differences in selection practices between the scientific domains. This is an interesting finding, as research fields differ in terms of dependency and uncertainty (Whitley, 2000). The SSH are characterized by more uncertainty than the natural and life sciences. Therefore one would expect contextual differences in quality attribution. However, this is not confirmed by our study. A possible explanation is that uncertainty may not only relate to scientific domains as a whole, but could also concern new and emerging fields within all domains.

Summarizing, our findings clearly indicate the contextuality of evaluation and decision-making. For improving transparency, quality and legitimacy of grant allocation practices, it would therefore be important to uncover more deeply the details of the *de facto* (implicit and explicit) applied criteria. As the selection procedure influences the evaluation of scientific talent, we suggest using a variety of procedures, instead of standardizing. The interview was found to have an important impact on the evaluation of the applicants. If communicative skills and self-confidence are decisive in this phase of the procedure, the selection outcomes will be biased towards these qualities, when all procedures would include interviews. Since no evident pool of talents could be identified based on the various scores, and as differences between granted and eventually rejected applications were small, a variety of procedures may result into the selection of a variety of talent.

Notes

1 The study was conducted in collaboration with the Netherlands Organisation for Scientific Research NWO. Cas Maessen, Mariken Elsen and two anonymous reviewers provided useful suggestions for improvement.
2 For a more elaborate literature review on the process of grant reviewing and group decision-making: van Arensbergen *et al*. (2012), Olbrecht and Bornmann (2010). For an elaborate review on peer review including the reviewing of scientific articles, see Bornmann (2011).
3 More precisely this is the quality, innovative nature and academic impact of the proposed research.
4 This is the intended societal, technological, economic, cultural or policy-related use of the knowledge to be developed over a period of 5–10 years.

5 From 2011 the Research Impact score will no longer be left out if it is detrimental to the total committee score, but will always be included in the calculation of the total committee score.
6 These are the following divisions: (1) Earth and life sciences; (2) chemistry; (3) mathematics, computer science and astronomy; (4) physics; (5) technical sciences; (6) medical sciences; (7) social sciences; (8) humanities. About 7% of the applications are cross-divisional.
7 We aggregated the scientific dimensions to domain level as follows: SSH=social sciences and humanities; STE=chemistry, mathematics, computer sciences and astronomy, physics, and technical sciences; LMS=earth and life sciences, and medical sciences.
8 Note that the applications are sent to different external reviewers, so generally reviewers are involved in the evaluation of a single application.
9 Since the data set is characterized by a large number of tied ranks, we use Kendall's tau instead of Spearman's rho.
10 Please note that also here lower scores correspond with higher numbers.
11 As the following results show, the stepwise method eliminates two variables since they do not contribute significantly to the model.
12 This is in line with the findings by Hodgson (1995), and contrasts with the findings of Bornmann *et al*. (2008).

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